SUMMARY · MK 4 OPERATIONS & SUPPLY CHAIN MANAGEMENT · WEEK 10

Statistical Quality Control

This week teaches one habit: compute first, conclude second. It starts with two kinds of variation (common cause, always present, vs assignable cause from outside the process), then the difference between specification limits (set by the designer) and process limits (set by the process), and the capability indices Cp and Cpk. The SPC part covers when to use attribute data (counted: p- and c-charts) or variable data (measured: $\bar{X}$- and R-charts, always as a pair), how to compute control limits with Chapter 13 examples, and how to read patterns (runs, trends, cycles) even when every point is inside the limits. Acceptance sampling judges a finished lot with two numbers (n and c), AQL and LTPD, producer's risk α and consumer's risk β, and the OC curve. The systems part covers the seven elements of a Quality Management System and Quality 4.0, then service quality through the five SERVQUAL gaps and the five RATER dimensions. The case is KAI Commuter: in control is not the same as capable.

Material

1 Two kinds of variation, and spec limits vs process limits

This week's ground rule
Bring the calculation, not an impression. Name the number that would change your mind.

A spec asks "is this part OK?". A control chart asks "has the process changed?".

Common cause (normal variation)

Many small sources, always there, predictable. Example: the pickle sits nearer the bun's edge, but it's there. **Adjusting a common-cause process adds variation.**

vs

Assignable cause (special variation)

One source from outside the process. Example: a worn tool, an untrained operator, a new subcontractor. **Ignore it and**

defects drift. A control chart prevents both mistakes.

The bearing example (Chapter 13): design 1.250 ± 0.005 in; process mean 1.250; σ = 0.002.

Limits	Set by	Range	Width
Specification (LSL–USL)	The designer	1.245 – 1.255	0.010 in
Process (±3σ)	The process itself	1.244 – 1.256	0.012 in
Six Sigma process (±6σ inside spec)	Improvement target	σ ≤ 0.005 ÷ 6 = 0.00083	±3σ only 0.005 in

What it means

The process is wider than the spec (0.012 > 0.010). Even perfectly centred, it will make parts the spec already rules out.

2 Process capability: Cp and Cpk

```
Cp = (USL - LSL) ÷ 6σ · Cpk = min(USL - μ, μ - LSL) ÷ 3σ
```

Cp

Would the spread *fit* inside the spec if the process were perfectly centred? Looks at width only.

vs

Cpk

Does it fit where the process *actually* sits? Penalises being off-target. Cpk ≤ Cp; equal only when centred.

±0.005 in tolerance	σ	Cp	Meaning
Bearing today	0.002	0.010 ÷ 0.012 = **0.83**	Below 1: can't meet spec even when centred
Just fits	0.00167	1.00	±3σ exactly fills the spec; about 0.27% outside

Capable	0.00125	1.33	A common minimum standard
Six Sigma	0.00083	2.00	±6σ fits; about 2 defects per billion when centred

A Cpk example: a capable process (σ = 0.00125) whose mean drifts to 1.251. Cpk = min(1.255 − 1.251; 1.251 − 1.245) ÷ (3 × 0.00125) = min(0.004; 0.006) ÷ 0.00375 = **1.07**. Cp stays 1.33, but the drift cuts real capability.

≡ **Lesson**

Halving σ cuts defects by orders of magnitude. That's why Six Sigma focuses on reducing variation, not just moving the mean.

3 SPC basics: attribute vs variable, and choosing the chart

3σ

The standard limit: 99.7% of points from an unchanged process fall inside.

25

Samples to collect before setting up a chart.

4–5

Units per sample for variable data; their means are near-normal.

1 in 370

Points outside 3σ by pure chance: the false-alarm budget.

! **Two reading rules**

A point outside = the process has changed. A point inside ≠ proof it hasn't. And *in control* = predictable, not necessarily *acceptable*.

	Attribute (count)	Variable (measure)
Measures	Good or bad (go/no-go). Over 10 lb = reject.	Actual values: 9.8 lb, not "heavy".
Charts	p-chart (fraction defective), c-chart (defects per unit)	$\bar{X}$-chart (centre) + R-chart (spread), always both

Limits	$\bar{p} \pm z\cdot\sqrt{(\bar{p}(1-\bar{p})/n)} \cdot \bar{c} \pm z\cdot\sqrt{\bar{c}}$	$\bar{\bar{x}} \pm A2\cdot\bar{R} \cdot D3\cdot\bar{R}$ to $D4\cdot\bar{R}$
Sample size	Large: expect the defect about twice. 1% → 200 units	Small: 4–5 units, taken quickly

Four design decisions behind any chart: sample size, number of samples, sampling frequency, and control limits (how many σ).

What does the data look like?	Chart	Question answered
Defective units in a fixed n	**p-chart**	Has the fraction defective changed?
Defects per unit of opportunity	**c-chart**	Has the defect count per unit changed?
Measurements (seconds, grams, mm)	**$\bar{X}$-chart + R-chart**	Has the mean moved? Has the spread widened?

Inside the limits is not all clear. Watch for patterns: points hugging a limit, *runs* (many points in a row on one side of the centre line), *trends* (steadily up or down) and *cycles* (repeated ups and downs).

> ! **Picking the wrong chart**
> The limits still look official but mean nothing. The data picks the chart, not preference.

4 p-charts and c-charts: worked examples

```
p-chart: p̄ = total defectives ÷ total inspected · σp = √(p̄(1 - p̄) ÷ n) · UCL/LCL = p̄ ± z·σp (LCL floored at 0)
```

Example 13.2: insurance claim forms (10 days × 300 forms, 91 defective).

1. $\bar{p} = 91 \div 3{,}000 =$ **0.0303**
2. $\sigma p = \sqrt{(0.0303 \times 0.9697 \div 300)} =$ **0.0099**

3 UCL = 0.0303 + 3 × 0.0099 = **0.0600**; LCL = 0.0303 − 0.0297 = **0.0006**

Solved Problem 2: 15 daily samples × 100 forms, 46 with errors, 95% limits (z = 1.96).

1 $\bar{p} = 46 \div 1{,}500 =$ **0.0307**

2 $\sigma p = \sqrt{(0.0307 \times 0.9693 \div 100)} =$ **0.017**

3 UCL = 0.0307 + 1.96 × 0.017 = **0.064**; LCL negative → **0**

4 Samples 6 (0.08) and 11 (0.07) break the UCL → **out of control**. Find the cause; don't widen the limit.

! **Same data, different verdict**
At z = 3, UCL = 0.0307 + 3 × 0.017 = 0.082 and nothing breaks. Choosing z is a decision about risk: tight limits = more false alarms; wide limits = more missed shifts.

```
c-chart (Poisson): σ = √c̄ · UCL/LCL = c̄ ± z·√c̄ (LCL floored at 0)
```

Example 13.3: average defects per unit $\bar{c} = 4 \rightarrow \sigma = \sqrt{4} = 2 \rightarrow$ UCL = 4 + 3 × 2 = **10**; LCL = 4 − 6 → **0** (counts can't be negative).

≡ **When to use a c-chart**
When you count defects on one "unit of opportunity": scratches per glass sheet, errors per report page, complaints per day, potholes per km of road.

5 $\bar{X}$-charts and R-charts

```
X̄-chart: UCL/LCL = x̿ ± A2 × R̄ · R-chart: UCL = D4 × R̄, LCL = D3 × R̄
```

The factors A2, D3, D4 are read from the table (Exhibit 13.7) for the sample size n. They assume a normal distribution.

n	A2	D3	D4	n	A2	D3	D4
2	1.88	0	3.27	7	0.42	0.08	1.92
3	1.02	0	2.57	8	0.37	0.14	1.86
4	0.73	0	2.28	9	0.34	0.18	1.82
5	**0.58**	**0**	**2.11**	10	0.31	0.22	1.78
6	0.48	0	2.00	15	0.22	0.35	1.65

D3 = 0 for n < 7 because the 3σ lower limit would fall below zero, so it's floored there. A range of zero is no signal for small samples.

Exhibits 13.8–13.9: 25 samples, n = 5, $\bar{\bar{x}}$ = 10.21, $\bar{R}$ = 0.60.

1. **$\bar{X}$-chart:** A2 × $\bar{R}$ = 0.58 × 0.60 = 0.348. UCL = 10.21 + 0.348 = **10.56**; LCL = 10.21 – 0.348 = **9.86**.
2. **R-chart:** UCL = 2.11 × 0.60 = **1.27**; LCL = 0 × 0.60 = **0**.

Reading the $\bar{X}$-chart

All 25 means are inside the limits. Which point would you chase? Most say sample 23 (hugs the LCL). Better: **the run at samples 13–17 above the mean**. A sustained shift says more than one low reading.

Reading the R-chart

Samples 18 and 21 hit a range of 1.00, invisible on the $\bar{X}$-chart. **Read R first:** if the spread is unstable, the $\bar{X}$ limits (computed from $\bar{R}$) can't be trusted.

Why always both

The mean can hold while the spread widens. $\bar{X}$ guards the centre, R guards the spread.

6 Acceptance sampling and the OC curve

Acceptance sampling judges a **lot that already exists** (not a running process). Every plan comes down to two numbers: **n** (sample size) and **c** (maximum defects allowed). Defects ≤ c → accept the lot; defects > c → reject it.

Producer's risk (α)

A *good* lot (≤ AQL) is rejected. The producer pays. **AQL** = the quality level counted as good.

vs

Consumer's risk (β)

A *bad* lot (≥ LTPD) is accepted. The consumer pays, downstream. **LTPD** = the quality level counted as bad.

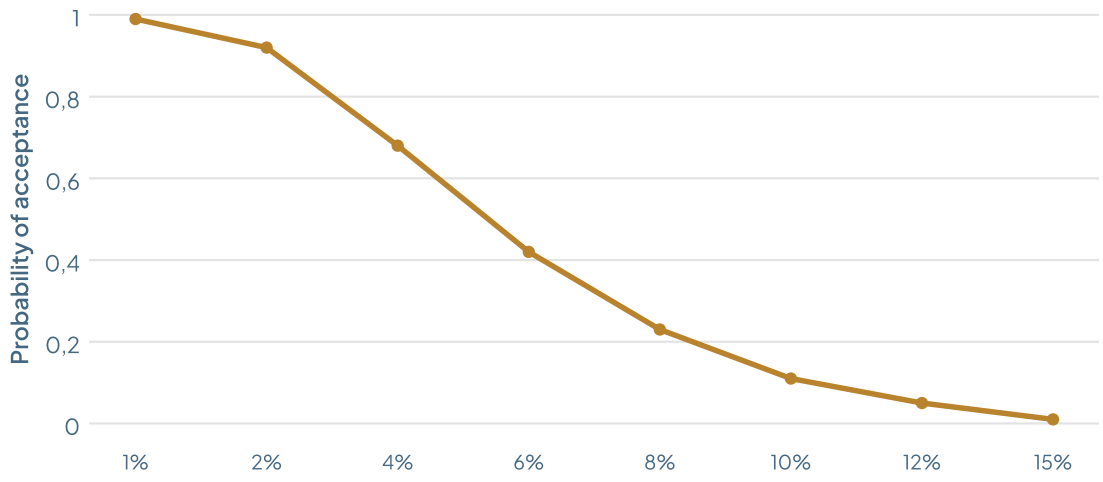
Remember

A plan tests the lot; it does not clean it. Defects in an accepted lot are still there.

When does inspection pay? When the loss if a defect is missed > the cost to inspect. Example: $10 loss per defect × 3% defect rate = **$0.30 per item**. If inspection costs less than $0.30 per item, it pays. **All-or-nothing rule:** if you already know the lot's quality, inspect all or none.

The **OC (operating characteristic) curve** shows the probability of accepting a lot by its true % defective. For **n = 50, c = 2** (binomial):

OC curve for n = 50, c = 2



A lot at 1.7% defective (AQL) is accepted 95% of the time (α = 0.05); a lot at 10.3% (LTPD) is still accepted 10% of the time (β = 0.10).

True % defective	1%	1.7% (AQL)	4%	6%	8%	10.3% (LTPD)	15%
P(accept)	0.99	**0.95**	0.68	0.42	0.23	**0.10**	0.01

So a lot as good as the AQL is accepted 95 times in 100 ($\alpha = 0.05$); a lot as bad as the LTPD still slips through 10 times in 100 ($\beta = 0.10$).

Two dials

Raise n → a steeper curve (sharper at telling good lots from bad). Raise c → the curve shifts right (looser).

7 The Quality Management System and Quality 4.0

A control chart is an instrument; the **QMS** acts on its reading. Seven elements decide what happens to a signal:

QMS element	Role
Management responsibility	Review and audit
Resources	People and infrastructure
Design control	Spec limits are set here
Process and production	Where SPC lives
Product surveillance	Complaints and monitoring
Corrective & preventive action	Fix, then verify
Change management	New process = new control limits

Surveillance spots it → Corrective action fixes it → Change management resets limits

Why the loop matters

Break any link and the chart still plots, but nothing changes. A chart can't tell you who owns the response, whether the fix was verified, or whether the change was authorised.

Quality 4.0: people, process, technology. What changes when data arrives continuously instead of weekly?

Process

Quality plans and non-conformance handling move into software; no hand transcription.

Technology

Connected products, data and analytics push data from the edge; no weekly paper report.

People

Culture, leadership, competency and visible quality KPIs. The ring most often skipped.

The analytics ladder: descriptive → diagnostic → predictive → prescriptive. A control chart is diagnostic; Quality 4.0 predicts before a limit is crossed.

≡ **An important caveat**

More data shows a bad process faster; it doesn't make it better. A stable-but-incapable process stays incapable at any sampling rate.

8 Service quality: the five SERVQUAL gaps and RATER

According to Parasuraman, Zeithaml and Berry, service quality is a **distance** between expectation and perception, not just a score.

Gap	Name	Meaning
1	Knowledge	The firm misreads what customers expect
2	Standards	Expectations are understood but never written into a standard
3	Delivery	The standard exists; delivery misses it
4	Communication	The promise is bigger than the delivery
5	Customer	Expected vs perceived: the sum of gaps 1–4

Customer expectations come from word of mouth, personal needs and past experience, plus the firm's external communications.

≡ **The slides' example (Gati)**

Gap 2 is the root (no SOP), gap 3 shows it, gap 4 makes it worse.

RATER: five dimensions customers judge on (ten original dimensions cut to the five that survived testing):

Dimension	Meaning	Slide note
Reliability	Deliver the promise accurately, every time	The only one SPC can measure
Assurance	Knowledge and courtesy that build trust	Competence you can see
Tangibles	Facilities, equipment, people, materials	Easiest to buy, weakest predictor
Empathy	Caring, individual attention	Where service recovery is won
Responsiveness	Willing to help, quickly	React fast when it goes wrong

! **Link to SPC**

SPC measures conformance to a spec; RATER measures how customers judge. Only Reliability fits on a control chart. A process can be *in control* and still lose on Assurance and Empathy.

Key takeaways

1. Common cause = normal variation (don't tamper); assignable cause = an outside source (find and fix it).
2. Spec limits come from the designer; process limits (±3σ) from the process. Bearing: process 0.012 > spec 0.010.
3. Cp = (USL − LSL) ÷ 6σ (fits if centred); Cpk = min(USL − μ, μ − LSL) ÷ 3σ (fits where it actually sits).
4. Bearing Cp = 0.83 (not capable); 1.33 = capable; 2.0 = Six Sigma (about 2 defects per billion).
5. A point outside = the process changed; inside ≠ all clear (runs, trends, cycles). In control ≠ acceptable.
6. Attribute (count): p-chart, c-chart. Variable (measure): $\bar{X}$-chart + R-chart, always both; read R first.
7. p-chart: $\bar{p} \pm z\sqrt{\bar{p}(1-\bar{p})/n}$. c-chart: $\bar{c} \pm z\sqrt{\bar{c}}$. $\bar{X}$: $\bar{\bar{x}} \pm A2\bar{R}$. R: $D3\bar{R}$ to $D4\bar{R}$.
8. Choosing z is a risk decision: z = 1.96 gives more false alarms; z = 3 more missed shifts.
9. Acceptance sampling: n and c; α (good lot rejected, AQL) vs β (bad lot accepted, LTPD); the OC curve.
10. Seven QMS elements; the loop surveillance → corrective action → change management. Quality 4.0: people, process, technology.
11. SERVQUAL: five gaps (knowledge, standards, delivery, communication, customer). RATER: reliability, assurance, tangibles, empathy, responsiveness.
12. KAI Commuter: stable at 5.25% (in control) but 75% above the 3% contract ceiling (not capable); capability comes after control.

Case studies

Case & worked answers

SOE case: KAI Commuter — in control, but good enough?

Setting: Juanda Station, Monday 2 March 2026, 6.40 a.m. The Managing Director needs one answer today: "Is our process still in control?" VP Operations Quality Rania Prameswari: "The right statistical answer may not be the useful one."

179.1 million
KAI Commuter riders in H1 2024 (82% of KAI Group)

1,100
Trips a day in Greater Jakarta (new timetable)

5.25%
Trips late (> 5 min) across 24 sampled weeks

≤ 3.00%
Contract ceiling, per line, per month

Trains **leave** on time (departure OTP > 99% for five years) but **arrive** late (arrival OTP never reached 97%; a 3.72-point gap in 2024). The delay happens between stations: on the line and at interchanges. On Rania's desk: 24 weeks of trip data (n = 500/week), 20 days of dwell time at Manggarai, 350 cause nominations from staff, and no check that the data is right.

Q1 · Is OTP still in control? On-time/late per trip with a fixed n = a **p-chart**.

1. $\bar{p} = 0.0525$; $\sigma_p = \sqrt{(0.0525 \times 0.9475 \div 500)} =$ **0.00997**
2. $z = 3$: UCL = 0.0525 + 0.0299 = **8.24%**; LCL = 0.0525 − 0.0299 = **2.26%**. No point crosses a limit.
3. But weeks 1–12 all sit below the centre line and weeks 13–24 all above: a **run of 12 points** signals a shift (the new timetable changed the process).
4. "In control" only means stable at 5.25%, which is **75% above the contract ceiling** (5.25 ÷ 3.00 = 1.75). Stable, but not capable.

Q2 · What if z = 1.96? UCL = 0.0525 + 1.96 × 0.00997 = **7.20%**; LCL = **3.30%**. Not one number about the process moved; only the risk decision changed. Tight limits = more false alarms; wide limits = more missed shifts. With contract penalties and safety at stake, which mistake costs more? And before arguing about z, check the data: some is typed in by staff who are judged on it.

Q3 · Dwell time at Manggarai. Measured data (seconds) → $\bar{X}$- and R-charts (n = 5: A2 = 0.577; D4 = 2.115; D3 = 0). One-sided capability with USL = 70 seconds:

```
Cpu = (USL - x̿) ÷ 3σ, with σ = R̄ ÷ d2 (d2 = 2.326 for n = 5)
```

The result is Cpu = 1.95, which passes any review, yet days 13–15 break the chart. Both are right: the index assumes a stable process, and the chart shows it isn't stable. **Capability only means something once the process is in control.** Fix first, then measure.

Q4 · SERVQUAL and the QMS. Pareto of 350 nominations: rolling stock 96 + infrastructure 74 + dwell time 58 = 228 = **65.1%**. Nobody named missing standards, training or procedures. That doesn't mean the problem doesn't exist; it may be invisible (gap 2). The cheap shortcut of lowering the promise or moving "late" from 5 to 7 minutes would improve tomorrow's numbers for free, but only cosmetically shrinks gap 4; the passenger experience (gap 5) stays the same.

A sample recommendation sequence: (1) validate the data (measure the measurement first); (2) recompute limits after the new timetable (change management); (3) stabilise Manggarai dwell time and tackle the top Pareto causes through corrective action; (4) only once stable, assess capability against the 3% contract ceiling; (5) don't redefine lateness.

Six takeaways

(1) In control ≠ capable. (2) Inside the limits ≠ all clear; 12 points in a row on one side is a signal. (3) Process change resets limits. (4) The data decides the chart. (5) Capability comes after control. (6) Measure the measurement first.

Worked examples

Problem 1 · Easy · **Cp and Cpk**

Bearing spec 1.250 ± 0.005 in. (a) σ = 0.002, centred: compute Cp. (b) After improvement σ = 0.00125, but the mean drifts to 1.251: compute Cp and Cpk.

1. `Cp = (USL - LSL) ÷ 6σ · Cpk = min(USL - μ, μ - LSL) ÷ 3σ`
2. (a) Cp = 0.010 ÷ (6 × 0.002) = 0.010 ÷ 0.012 = 0.83
3. (b) Cp = 0.010 ÷ (6 × 0.00125) = 0.010 ÷ 0.0075 = 1.33
4. (b) USL − μ = 1.255 − 1.251 = 0.004; μ − LSL = 1.251 − 1.245 = 0.006
5. (b) Cpk = 0.004 ÷ (3 × 0.00125) = 0.004 ÷ 0.00375 = 1.07

Answer: (a) Cp 0.83 (not capable). (b) Cp 1.33 but Cpk 1.07: the drift cuts real capability.

Problem 2 Medium **p-chart limits**

15 daily samples of 100 claim forms, 46 with errors in total. Compute p-chart limits at z = 1.96 and z = 3. Sample 6 shows 0.08 and sample 11 shows 0.07. Is the process in control?

1. `p̄ = Σ defectives ÷ Σ n · σp = √(p̄(1 - p̄) ÷ n) · limits = p̄ ± z·σp`
2. $\bar{p} = 46 \div 1{,}500 = 0.0307$
3. $\sigma_p = \sqrt{(0.0307 \times 0.9693 \div 100)} = 0.0172$
4. z = 1.96: UCL = 0.0307 + 0.0338 = 0.064; LCL negative → 0
5. z = 3: UCL = 0.0307 + 0.0517 = 0.082; LCL → 0

Answer: At 95% (UCL 0.064), samples 6 and 11 break out → out of control; find the cause. At 3σ (UCL 0.082), nothing breaks.

Problem 3 Easy **c-chart limits**

A call centre averages 9 complaints a day. Compute the 3σ c-chart limits. If there are 19 complaints today, what does it mean?

1. $\sigma = \sqrt{\bar{c}} \cdot UCL/LCL = \bar{c} \pm 3\sqrt{\bar{c}}$
2. $\sigma = \sqrt{9} = 3$
3. $UCL = 9 + 3 \times 3 = 18$
4. $LCL = 9 - 9 = 0$
5. $19 > 18 \rightarrow$ outside the limit

Answer: Limits 0–18. 19 complaints = an assignable-cause signal (e.g. a system outage); find the cause that day.

Problem 4 Medium **$\bar{X}$- and R-charts**

25 samples of n = 5: $\bar{\bar{x}}$ = 10.21 and $\bar{R}$ = 0.60. From the table: A2 = 0.58; D3 = 0; D4 = 2.11. Compute the $\bar{X}$- and R-chart limits. Sample 18 has a range of 1.00; what does it mean?

1. $\bar{X}$: $\bar{\bar{x}} \pm A2 \cdot \bar{R}$ · R: $D3 \cdot \bar{R}$ to $D4 \cdot \bar{R}$
2. A2 × $\bar{R}$ = 0.58 × 0.60 = 0.348
3. $\bar{X}$: UCL = 10.21 + 0.348 = 10.56; LCL = 10.21 − 0.348 = 9.86
4. R: UCL = 2.11 × 0.60 = 1.27; LCL = 0
5. Range 1.00 < 1.27 → still inside the R limits, but well above $\bar{R}$ 0.60; check for a widening-spread pattern

Answer: $\bar{X}$: 9.86–10.56; R: 0–1.27. The 1.00 range is invisible on the $\bar{X}$-chart: why $\bar{X}$ and R are always used together.

Problem 5 Medium **One-sided Cpu (dwell time)**

Illustrative figures. Station dwell time: $\bar{\bar{x}}$ = 55 seconds, $\bar{R}$ = 6 seconds, n = 5 (d2 = 2.326). Upper spec limit 70 seconds. Compute Cpu. What must hold for it to be trusted?

1. $\sigma = \bar{R} \div d2 \cdot Cpu = (USL - \bar{\bar{x}}) \div 3\sigma$
2. $\sigma = 6 \div 2.326 = 2.58$ seconds
3. $Cpu = (70 - 55) \div (3 \times 2.58) = 15 \div 7.74 = 1.94$
4. Condition: the $\bar{X}$- and R-charts show the process is in control

Answer: Cpu ≈ 1.94 (looks very good). But if some days break the chart, the process isn't stable and Cpu misleads: stabilise first, then measure capability.

Problem 6 Easy **When inspection pays**

Each defective part that reaches the customer costs $10. The defect rate is 3%. Inspection costs $0.25 per item. Is 100% inspection worth it? What if the defect rate falls to 1%?

1. Expected loss per item = cost of a defect × defect rate
2. 3%: 10 × 0.03 = $0.30 > $0.25 → inspection pays
3. 1%: 10 × 0.01 = $0.10 < $0.25 → it doesn't

Answer: At 3% inspect all (saves $0.05/item); at 1% don't inspect. The all-or-nothing rule.

Practice questions

Try answering first. The answer key is below each question.

1. A worn tool, an untrained operator or a new subcontractor are examples of…

A. Common cause
B. Assignable cause
C. Spec limits
D. Consumer's risk

Answer: B. One source from outside the process; find and fix it.

2. What happens if you keep adjusting a process with only common-cause variation?

A. Variation falls
B. Variation increases
C. Cpk rises
D. No effect

Answer: B. Tampering with a stable process adds variation.

3. Bearing: spec 1.250 ± 0.005; σ = 0.002. Its Cp is…

A. 0.42
B. 0.83
C. 1.00
D. 1.33

Answer: B. 0.010 ÷ (6 × 0.002) = 0.83: not capable even when centred.

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4. How Cpk differs from Cp…

A. Cpk looks at width only
B. Cpk accounts for where the mean sits and penalises drift
C. Cpk is always larger
D. Cpk is for attribute data

Answer: B. Cpk ≤ Cp; equal only when centred.

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5. An "in control" process means…

A. It meets spec
B. Stable and predictable, not necessarily acceptable
C. It has no defects
D. Cp > 1.33

Answer: B. KAI is stable at 5.25% but the contract ceiling is 3%.

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6. Counting scratches per glass sheet uses a…

A. p-chart
B. c-chart
C. $\bar{X}$-chart
D. R-chart

Answer: B. Defects per unit of opportunity = c-chart (Poisson).

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7. The fraction of late trips out of 500 a week uses a…

A. p-chart
B. c-chart
C. $\bar{X}$-chart + R-chart
D. OC curve

Answer: A. Each unit on-time/late with fixed n = attribute fraction data.

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8. Example 13.2: 91 defectives in 3,000 forms (n = 300). The 3σ UCL is…

A. 0.0303
B. 0.0402
C. 0.0600
D. 0.0900

Answer: C. $\bar{p} = 0.0303$; $\sigma p = 0.0099$; UCL = $0.0303 + 3 \times 0.0099 = 0.0600$.

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9. Solved Problem 2: at z = 1.96 two samples break the UCL; at z = 3 none do. The lesson…

A. Always use z = 3
B. Choosing z is a decision about false alarms vs missed shifts
C. The data is wrong
D. A p-chart doesn't fit

Answer: B. Same data, different verdict. Only the risk tolerance changed.

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10. A c-chart with $\bar{c} = 4$. Its 3σ UCL is…

A. 6
B. 8
C. 10
D. 16

Answer: C. $\sigma = \sqrt{4} = 2$; UCL = $4 + 3 \times 2 = 10$; LCL = 0.

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11. n = 5, $\bar{\bar{x}} = 10.21$, $\bar{R} = 0.60$, A2 = 0.58. The $\bar{X}$-chart UCL…

A. 10.27
B. 10.56
C. 10.81
D. 11.48

Answer: B. $10.21 + 0.58 \times 0.60 = 10.21 + 0.348 = 10.56$.

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12. Why is D3 = 0 for n < 7?

A. The range can't be computed
B. The 3σ lower limit would be negative, so it's floored at zero
C. Small samples have no variation
D. It's a table error

Answer: B. A zero range is no signal for small samples.

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13. All $\bar{X}$ points are inside the limits, but samples 13–17 are all above the mean. What's most worth chasing?

A. Nothing, all clear
B. The run at 13–17, because a sustained shift is a signal
C. Only the lowest point
D. Widen the limits

Answer: B. Inside the limits ≠ all clear; watch for runs, trends, cycles.

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14. Why read the R-chart before the $\bar{X}$-chart?

A. R-charts are easier
B. $\bar{X}$ limits are computed from $\bar{R}$; if the spread is unstable, they can't be trusted
C. $\bar{X}$-charts don't matter
D. An admin rule

Answer: B. The mean can hold while the spread widens.

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15. A good lot (≤ AQL) is rejected. This is…

A. Consumer's risk β
B. Producer's risk α
C. LTPD
D. Capability

Answer: B. The producer pays. β = a bad lot (≥ LTPD) is accepted.

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16. On an OC curve, raising the sample size n…

A. Shifts the curve right
B. Makes the curve steeper
C. Removes β risk
D. Raises the AQL

Answer: B. Raising c shifts the curve right.

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17. A $10 loss per missed defect, 3% defect rate. Inspection pays if its cost per item is…

A. Above $3
B. Below $0.30
C. Below $10
D. Any amount

Answer: B. $10 × 3% = $0.30 per item.

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18. In a QMS, "new process = new control limits" is the role of…

A. Design control
B. Change management
C. Product surveillance
D. Resources

Answer: B. That's why KAI must recompute limits after the new timetable.

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19. Per the slides, continuously flowing quality data (Quality 4.0)…

A. Makes processes capable automatically
B. Shows a bad process faster but doesn't make it better
C. Replaces the QMS
D. Removes the need for SPC

Answer: B. Stable-but-incapable stays incapable at any sampling rate.

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20. Expectations are understood but never written into a standard. This is SERVQUAL gap…

A. 1
B. 2
C. 3
D. 4

Answer: B. Gap 1 = misread expectations; 3 = delivery misses; 4 = promise exceeds delivery.

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21. The RATER dimension the slides call easiest to buy but the weakest predictor is…

A. Reliability
B. Assurance
C. Tangibles
D. Empathy

Answer: C. Facilities and equipment are easy to buy but don't drive customer judgements much.

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22. KAI Commuter: $\bar{p}$ = 5.25%, n = 500. The UCL at z = 3 is about…

A. 6.2%
B. 7.2%
C. 8.2%
D. 10.5%

Answer: C. $\sigma p = \sqrt{(0.0525 \times 0.9475 \div 500)} = 0.00997$; UCL = 5.25% + 2.99% = 8.24%. At z = 1.96: 7.20%.

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23. Manggarai dwell-time Cpu is 1.95, yet days 13–15 break the chart. The right explanation…

A. Cpu was miscalculated
B. Capability indices assume stability; capability only means something once in control
C. The chart is wrong
D. Just use a p-chart

Answer: B. Fix first, then measure capability.

Sources

- P3MD MK4 Week 10 — Statistical Quality Control (OSCM); Jacobs & Chase, Operations and Supply Chain Management, Ch. 13
- Parasuraman, Zeithaml & Berry — SERVQUAL; Scilife — Quality Management System; Quality 4.0 People, Process and Technology Wheel